A Geometric Approach to Integration

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1 Introduction

This piece of writing aims to introduce the fabulous idea of measuring a mathematical object as general as a manifold, an object so abstract that I would rather put it off until more serious texts. The point is not to rashly bombard the reader with axioms that seem to come from nowhere, but to study familiar examples in low dimensions with care and derive what we minimally require in the measurement of anything based on the same approach.

The essence of measurement consists of approximation and accumulation, which, for people versed in calculus, are also essential to calculus. Indeed, from a higher measure-theoretic perspective, this is not just an analogy; the two are nearly identical. We will take a tour of low-dimensional scenarios to introduce the Jacobian, study measurement in the linear world, and finally assemble everything by integration.

One can hardly study (real) numbers without the number line. This is when we realize that coordinates, or what we are most accustomed to, Cartesian coordinates, are fundamental in the study of geometric spaces. But elementary as they are, coordinates are a man-made apparatus nonexistent in the physical world. Only after emancipation from these artificial tuples at some point in the theory can we finally convince ourselves that the geometry we've been contriving is still physical. The introduction of coordinates should then be regarded as reluctant, and their disposal as satisfactory.

2 From Derivative to Differential

To measure anything, our intuition leads us to cut everything into small pieces from the very beginning. Imagine two copies of the real line, temporarily denoted by $\mathbb{R}_1$, $\mathbb{R}_2$, and a smooth map f from the former to the latter, that is, a rule assigning something in $\mathbb{R}_2$

to every single item in $\mathbb{R}_1$, with the output depending smoothly on the input. You may find it a weird model to visualize functions if you are comfortable with graphs, but this is where higher dimensionality comes into play, so it is more suitable for our discussion. We aim at measuring how far the output goes if the input runs through a designated connected region, or an interval $I \subset \mathbb{R}_1$.

The cutting begins. If we observe how the output varies as the input varies, it most likely carries no pattern. But if the input varies a tiny bit, or infinitesimally, the output does have a pattern: constancy in velocity. There is no point in arguing whether this pattern is applicable generally, since that is what we mean by smoothness in the first place. And as the input also varies only a little, we may assume both points to have constant velocity at this moment.

The previous process is formally called the linearization of the map f at any given point. To measure how much the output moves with the input, we consider the ratio of their velocities. At any point x whose vicinity is contained in I, or at any interior point of the interval, we define the ratio of the velocity of $f(x)$ to that of x, in the infinitesimal sense, to be the derivative of f at x, with the remarkable fact that this definition is independent of the chosen movement. After multiplying both velocities by a tiny time interval, this becomes the ratio of their infinitesimal displacements.

This observation makes it possible to compute the tiny length in the output space $\mathbb{R}_2$ as the corresponding one in $\mathbb{R}_1$ scaled by the derivative, which generally varies from point to point. But strictly speaking, to really measure the length, one needs non-negativity, so the derivative is often conveniently placed inside an absolute value to eliminate the impact of directions, or orientations.

In modern mathematical language, the derivative is denoted and defined by

$$f'(x) = \lim_{y \to x} \frac{f(y) - f(x)}{y - x},$$

which is accepted by any geometer through the intuition exactly illustrated as above.

By running through this process, one finds that it doesn't matter to replace the output space $\mathbb{R}_2$ with any higher-dimensional analogue, such as a Euclidean space, so long as the notions of velocity and displacement are generalizable. But with Cartesian coordinates as a tool at hand, we merely use multiple numbers, instead of a single one, to describe the output, so each output $f(x)$ can be written as a tuple $(f_1(x), \ldots, f_n(x))$, if the output space has dimension n. The derivative is defined using exactly the same formula as in the one-dimensional case, which is well-defined because the numerator is a vector and the denominator is a number (scalar). The absolute value of the derivative should be interpreted as the length of the derivative as a vector, defined in the Pythagorean sense.

There is finally an unexpected deviation when the input space $\mathbb{R}_1$ becomes multi-dimensional. While dividing a number, or even a vector, by a number is to scale the former object (direction unchanged or reverted), dividing anything by a vector sounds ridiculous. Indeed, I personally have so far not witnessed any division by a vector in any recondite branch of the subject. The good news is, we have an awesome replacement: if division doesn't make sense, why do we have to divide in the first place? Returning to the easiest case: why not define the map from input velocities to corresponding output velocities, instead of their quotients, as the derivative? It is only the one-dimensionality of the spaces in question that makes this ratio valid. The true story is that for higher dimensions, the notion of derivative splits into two: derivative and differential, the former of which is to be induced by the latter.

Instead of grappling with dividing a vector, we proceed to regard the smooth map f from U to $\mathbb{R}^n_2$ as a physical movement, where U is a region in $\mathbb{R}^m_1$. At any interior point $x \in U$, consider the infinitesimal movement of $f(x)$ with respect to that of x. As hinted above, one may define a map ∇f, called the differential of f at x, sending the velocity at x to that of $f(x)$. As usual, smoothness ensures that the definition is independent of the selected curve of motion. To better describe velocities, we regard all possible velocities at x as vectors, forming a vector space T_xU, called the tangent space at this point. You may think of vectors as literal arrows rooted there, satisfying the well-known rules of force in high school physics. Using this abstraction, the differential, or the Jacobian, which I prefer for better distinction and easier recollection of the mathematician, is a map from T_xU to $T_{f(x)}\mathbb{R}^n_2$.

The single most significant implication of smoothness, corresponding to the pattern of infinitesimally constant velocity of the simplest case, is the linearity of ∇f, namely, it respects vectorial operations: denoting the image of v by $\nabla_v f$, called the directional derivative of f along v, we have

$$\nabla_{av+bw} f = a\nabla_v f + b\nabla_w f$$

for scalars a, b. This crucial revelation opens up a linear world for calculus, establishing a fascinating arena for linear algebra to demonstrate its capability.

3 Gram Matrix: the Metric Ruler

The major difference for higher-dimensional Euclidean spaces is not just an explosive number of directions to choose from, but an otherworldly upgrade of our ruler. As the input point has more freedom to move around, so does the output, yielding a sophisticated object of higher dimensions that cannot be measured by the naive linear ruler. We say that this object, the image, is parametrized by the input. This may be interpreted as reading the information of a curvy object from coordinates in a flat region, like using latitudes and longitudes to determine points on the sphere, except the two poles. It is due to the arbitrary dimensionality of the output space, apart from the intrinsic information of the image, that we prefer not to read off the coordinates there.

To measure such an advanced object with the respective advanced ruler, and continuing the infinitesimal study using the Jacobian, we really need to return to the case where all infinitesimal studies can be handled directly because of its unique structure: vector spaces with linear structure. For this part, I would require the reader to have at least some superficial prerequisites in linear algebra.

Since we measure objects of dimension one by lengths, objects of dimension two by areas, and objects of dimension three by volumes, it would be out of the question and impossible to give new names to all the remaining measurements of an exotic nature. So why not call them all "measure" for uniformity? An incredible fact and the intrinsic reason why vector spaces are simple is, without doubt, that there is a basis that spans everything through linear combinations, with the number of basis vectors exactly equal to the dimension. Nobody who has heard of Cartesian coordinates representing the plane as made up of pairs of numbers is alien to this notion. Courtesy of this simplicity, it suffices to consider the oblique cube enclosed by any basis; we do have a name, parallelotope, for such objects in arbitrary dimensions, generalized from parallelograms in two dimensions and parallelepipeds in three. Formally speaking, it is described by the set

$$P = \left\{ \sum_{i=1}^{k} a_i v_i : 0 \leq a_i \leq 1, \forall\, i \right\}$$

if $\{v_i\}$ is the spanning set. This description already parametrizes the parallelotope by the unit cube $[0,1]^k = \left\{ (a_i)_{i=1}^k : 0 \leq a_i \leq 1, \forall\, i \right\}$, which is the parallelotope spanned by the standard (orthonormal) basis $\{e_i\}$ and has unit measure by convention. So to obtain the measure of P, one should study how the measurement is distorted under any change of basis, or the particular change of basis $\{e_i\} \to \{v_i\}$.

It would spoil all the fun to reveal the desired formula at the beginning, as it is the process of speculation and verification of this, and any, formula that makes mathematics appealing and rewarding. Instead of deriving the formula, let's first investigate what properties the measure of P depends on. Under rotation, reflection, or any transformation preserving both lengths and angles, this measure should stay invariant. We call transformations with this property orthogonal ones. Recall that the inner, or dot, product of vectors, in the traditional sense, encodes the lengths of these vectors and the angle between them in terms of its cosine. If we know the value of the inner product $\langle v, w \rangle$ as well as the lengths $|v| = \sqrt{\langle v, v \rangle}$, $|w| = \sqrt{\langle w, w \rangle}$, we know the angle between them and thus can derive the area of the spanned parallelogram. In other words, what we minimally need to know is all possible inner products within a set of vectors to fix their relative positions, without any care for the orientation. The measure of P, completely determined by the relative positions, should then depend on the values $\langle v_i, v_j \rangle$ and nothing else.

The parallelogram, in dimension two, is of particular importance, as it is the lowest nontrivial case in question. Denoting the vectors by v, w and the angle in between by θ, the area A is known to be $|v|\,|w| \sin\theta$, which is surprisingly similar to the inner product $\langle v, w \rangle = |v|\,|w| \cos\theta$. Combining this with the trigonometric identities, we easily derive the relation

$$A^2 = |v|^2\,|w|^2 - \langle v, w \rangle^2 = \det \begin{pmatrix} \langle v, v \rangle & \langle v, w \rangle \\ \langle w, v \rangle & \langle w, w \rangle \end{pmatrix},$$

where the determinant immediately betrays its validity in the one-dimensional analogue, $|v|^2 = \det(\langle v, v \rangle)$. Only now can we effortlessly and boldly speculate the higher-dimensional analogue of the measure of P as

$$\operatorname{Vol}(v_1, \ldots, v_k)^2 = \det \begin{pmatrix} \langle v_1, v_1 \rangle & \cdots & \langle v_1, v_k \rangle \\ \vdots & \ddots & \vdots \\ \langle v_k, v_1 \rangle & \cdots & \langle v_k, v_k \rangle \end{pmatrix} = \det(\langle v_i, v_j \rangle)_{1 \leq i,j \leq k}.$$

Now the fun part is over: what remains is the tedious process of verifying, first, that the formula is invariant under orthogonal and other transformations known a priori to preserve the measure and, second, that the formula is true in the simplest scenarios, a task on which I would suggest that motivated and diligent readers pause and work.

For people familiar with, or acquainted with, tensors, there is an elegant algebraic approach I did not discover until recently. We formulate the parallelotope as an alternating product $v_1 \wedge \cdots \wedge v_k$, since the object itself, apart from its measurement, does possess orientation properties, and formulate the measurement as some sort of norm induced by a natural inner product in the space of alternating products. By regarding alternating products as the alternation of tensor products, the latter being characterized by mere multilinearity, we inherit from tensor products a natural inner product for alternating products. In symbols, and for simplicity, in dimension two, it naturally holds that

$$\langle v_1 \otimes v_2, w_1 \otimes w_2 \rangle = \langle v_1, w_1 \rangle \langle v_2, w_2 \rangle ,$$

so under the alternation operation $v \wedge w = \frac{1}{2}(v \otimes w - w \otimes v)$, we end up with the norm squared defined by

$$|v \wedge w|^2 = \langle v \wedge w, v \wedge w \rangle = \frac{1}{2} \det \begin{pmatrix} \langle v, v \rangle & \langle v, w \rangle \\ \langle w, v \rangle & \langle w, w \rangle \end{pmatrix},$$

which, up to a constant, coincides with A^2 in the above discussion.

The matrix $(\langle v_i, v_j \rangle)_{1 \le i,j \le k}$ is known as the Gram matrix of the set $\{v_i\}$, encoding all the metric information as we have shown.

4 The Ubiquitous Parametric Integral

Though geometers are reluctant to use coordinates, we are bound to use them to compute specific examples, as there are truly few of them that possess any coordinate-free expression. The Jacobian ∇f as a linear map should have a matrix representation with respect to the standard bases of the input and output spaces. As we have seen, the Jacobian acts by taking the directional derivative; to obtain the i-th column, we consider its action on the i-th standard basis vector of the tangent space and get $\nabla f(e_i) = \nabla_{e_i} f = \partial_i f$, called the partial derivative of f in this direction. Putting these together, the Jacobian ∇f takes the form

$$\begin{pmatrix} | & \cdots & | \\ \partial_1 f & \cdots & \partial_m f \\ | & \cdots & | \end{pmatrix} = \begin{pmatrix} \partial_1 f_1 & \cdots & \partial_m f_1 \\ \vdots & \ddots & \vdots \\ \partial_1 f_n & \cdots & \partial_m f_n \end{pmatrix} = (\partial_j f_i)_{1 \le i \le n, 1 \le j \le m},$$

whose columns record where the standard basis $\{e_i\}$ of $T_x U$ is sent in $T_{f(x)}\mathbb{R}^n_2$, written in the coordinates of $\mathbb{R}^n_2$.

On the other hand, the Gram matrix also possesses a coordinate representation. Denoting A as a linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^n$ whose matrix in the standard basis of the two spaces is denoted by A as well, the images of the standard basis vectors $\{e_i\}$ of $\mathbb{R}^m$ are, as with the Jacobian, precisely the columns of A, $\{v_1, \ldots, v_m\}$. Recall that the traditional inner product $\langle v, w \rangle$ can be neatly expressed by the matrix product $v^\top w$ or $w^\top v$, which has been viewed as one of the motivations to define general matrix multiplication the way we currently use it. Hence, by concatenating this structure vertically and horizontally, the Gram matrix can be written as

$$\begin{pmatrix} \langle v_1, v_1 \rangle & \cdots & \langle v_1, v_m \rangle \\ \vdots & \ddots & \vdots \\ \langle v_m, v_1 \rangle & \cdots & \langle v_m, v_m \rangle \end{pmatrix} = \begin{pmatrix} - & v_1^\top & - \\ \vdots & \vdots & \vdots \\ - & v_m^\top & - \end{pmatrix} \begin{pmatrix} | & \cdots & | \\ v_1 & \cdots & v_m \\ | & \cdots & | \end{pmatrix} = A^\top A,$$

which is indeed a much more amicable form in the verification of the measure of P discussed in the previous section, as under any orthogonal transformation Q, which has determinant ± 1, the expression $\det(A^\top A)$ is easily seen to be invariant:

$$\det((QA)^\top (QA)) = \det(A^\top Q^\top QA) = \det(A^\top A).$$

Combining this algebraic observation with the Jacobian, it holds that the expression $\sqrt{\det((\nabla f)^\top \nabla f)}$ is the infinitesimal ratio between the measures of a parallelotope in $T_{f(x)}\mathbb{R}_2^n$ and the corresponding one in $T_x U$ under the map f. If, informally, the latter is denoted by $\mathrm{d}x_1 \cdots \mathrm{d}x_m$, in the spirit of multiplying the infinitesimal edges $\mathrm{d}x_i$ along each coordinate direction, the former measure, the one we wish to compute, equals $\sqrt{\det((\nabla f)^\top \nabla f)}\,\mathrm{d}x_1 \cdots \mathrm{d}x_m$. Finally, the microscopic picture accumulates into a global entity, at whose appearance, however, our intuition collapses; its measure can be written as the parametric integral

$$\int_U \sqrt{\det((\nabla f)^\top \nabla f)}\,\mathrm{d}x_1 \cdots \mathrm{d}x_m,$$

where we often rely on a cubical U and a simple $\sqrt{\det((\nabla f)^\top \nabla f)}$ to derive a numerical result by hand, and resort to the computer for general integral computations.

In practice, we usually start with a tortuously ugly geometric object, and then figure out parametrizations f to pull patches of this object back to some flat regions U smoothly, finally pasting the results together. We have illustrated this approach with an ambient space $\mathbb{R}_2^n$, but that turns out to be omissible in the more abstract theory of differentiable and Riemannian manifolds, so long as the object is locally parametrizable and carries a metric structure.

Ubiquitous as it is, all kinds of integrals in calculus can more or less be reduced to this prototypical type: the parametric integral.